Problem solving from lecture slides for epidemiological measures

To calculate the attack rate of the cholera outbreak in Haiti between January and October 2010, you can use the following formula:

Attack Rate = (Number of Cases / Population at Risk) x 1000

In this case:

Number of Cases = 470,000
Population at Risk = 8,500,000

Attack Rate = (470,000 / 8,500,000) x 1000
Attack Rate = (0.0552941176) x 1000
Attack Rate ≈ 55.29

So, the attack rate of the cholera outbreak in Haiti between January and October 2010 is approximately 55.29 cases per 1,000 people at risk.

Problem 2

To calculate the cumulative incidence of Sudden Infant Death Syndrome (SIDS) during the first year in a population of 1,000 live births with 10 cases of SIDS, you can use the following formula:

Cumulative Incidence = (Number of New Cases / Total Population at Risk) x 1000

In this case:

Number of New Cases = 10 cases of SIDS
Total Population at Risk = 1,000 live births

Cumulative Incidence = (10 / 1,000) x 1000
Cumulative Incidence = (0.01) x 1000
Cumulative Incidence = 10

So, the cumulative incidence of Sudden Infant Death Syndrome (SIDS) during the first year in this population is 10 cases per 1,000 live births.